Applied Databases

Higher Diploma in Science in Data Analytics

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1 Description

This document describes the final project specification for the Applied Databases module.

2 Marks

This project is worth 60% of the marks for the module.

2.1 Marking Scheme

70% of the marks will be awarded for implementing the functionality described in this document.

30% of the marks will be awarded for innovation and extra functionality in the Python App.

Please describe your innovation (if any) in a document entitled *innovation.doc* or *innovation.pdf* which should be stored in root folder of your project.

If you are using any extra Python packages in your project as part of your innovation, please note these will not be installed on the machine your programme is being tested on. So, you should preferably have these packages installed automatically by your programme, or at a minimum give the exact command needed to manually install them in the Innovation document.

Everything will be tested on the Virtual Machine (VM), so if you are using on your own laptop, you should still ensure that everything works on the VM.

NOTE: You may be invited to an MS Teams meeting for a viva explanation of any or all parts of your submission.

A functioning application that a student cannot clearly explain may incur a penalty proportionate to the extent of that lack of understanding

2.1.1 Plagiarism

Plagiarism will be dealt with in accordance with the university's Student Code.

3 Submission

Your Python app should be compressed into a zip file called *GXXXXXXXX.zip* (where GXXXXXXXX is your student number) and uploaded to Moodle before **Wednesday May 13th 2026 at 11:59pm**.

Your project should be developed over time in GitHub, with each commit reflecting the changes you have made.
It is not sufficient to simply have one or two commits, regular commits with comments over time is required.
A file entitled *GitLink.pdf* (or *GitLink.txt*) must be included in the root folder of your application.
Ensure your repository is publicly accessible, or if you'd like to keep it private add me as a collaborator (GitHub Profile: gerardharrisonatu, email:Gerard.Harrison@atu.ie).

3.1 Python Application

The main file of your application should be called *main.py.*

3.1.1 Databases

The application uses the following databases:

- MySQL
  Download *appdbproj.sql* from Moodle and import it into MySQL, which will result in a database called *appdbproj*.
  This is a conference management database containing a number of interrelated tables showing who attended what session and where.

- Neo4j
  Download *appdbprojNeo4j.json* from Moodle and import into a Neo4j database called *appdbprojNeo4j*.

  This database shows "connections" between attendees. Any attendee (that is in the MySQL database) can have a *CONNECTED_TO* relationship with any other attendee.
  The direction of the *CONNECTED_TO* relationship is unimportant. In this example, attendee 108 is *CONNECTED_TO* attendee 116, but attendee 116 is also *CONNECTED_TO* attendee 108.

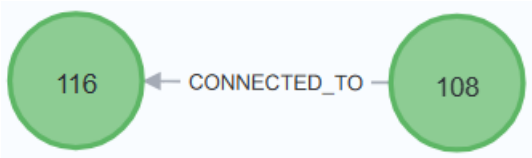

Figure 1 A CONNECTED_TO relationship

NOTE: You can assume that any attendee in the Neo4j database will <u>always</u> exist in the MySQL database.

3.1.2 Main Menu

Write a python application that displays a main menu as follows:

```
Conference Management
---------------------

MENU
====
1 - View Speakers & Sessions
2 - View Attendees by Company
3 - Add New Attendee
4 - View Connected Attendees
5 - Add Attendee Connection
6 - View Rooms
x - Exit application
Choice: |
```

Figure 2 Main Menu

3.1.3 1 (View Speakers & Sessions)

The user is asked to enter a speaker name (or part thereof):

```
Conference Management
---------------------

MENU
====
1 - View Speakers & Sessions
2 - View Attendees by Company
3 - Add New Attendee
4 - View Connected Attendees
5 - Add Attendee Connection
6 - View Rooms
x - Exit application
Choice: 1

Enter speaker name : dr
```

Figure 3 Enter Speaker Name

For each speaker that has those letters in his/her name the following details are shown:

- The speaker's name
- The title of the session the speaker is giving
- The name of the room the session is being held in.

```
Conference Management
---------------------

MENU
====
1 - View Speakers & Sessions
2 - View Attendees by Company
3 - Add New Attendee
4 - View Connected Attendees
5 - Add Attendee Connection
6 - View Rooms
x - Exit application
Choice: 1

Enter speaker name : dr
Session Details For :  dr
---------------------------------------------
Dr. Niamh Burke  |  Modern Data Pipelines  |  Main Hall
Dr. Niamh Burke  |  Customer 360 with SQL  |  Main Hall
Dr. Sara Khan  |  AI in Healthcare Operations  |  Main Hall
```

Figure 4 Speaker sessions

When all data has been shown, the user is returned to the main menu.

If no speaker matches the search string the following message is shown, and the user is returned to the main menu.

```
Enter speaker name : xyz
Session Details For :  xyz
---------------------------------------------
No speakers found of that name
```

Figure 5 No speakers match search string

3.1.4 2 (View Attendees by Company)

The user is asked to enter a company ID:

```
Conference Management
---------------------

MENU
====
1 - View Speakers & Sessions
2 - View Attendees by Company
3 - Add New Attendee
4 - View Connected Attendees
5 - Add Attendee Connection
6 - View Rooms
x - Exit application
Choice: 2

Enter Company ID : 2
```

Figure 6 Enter Company ID

When a valid (numeric) company ID is entered, the company name is shown and the following details are shown for each attendee from that company:

- The name of each attendee
- The date of birth of each attendee
- The title of the session the attendee attended
- The name of the speaker at the session the attendee attended
- The name of the room the session was held in.

```
Enter Company ID : 2
CloudSprint  Attendees
Cian Roche  |  1989-9-8  |  Cloud Cost Optimisation  |  Ruth Collins  |  2025-05-14  |  Cloud Suite
Cian Roche  |  1989-9-8  |  FinTech Risk Signals  |  Marta Silva  |  2025-05-13  |  Executive Lounge
Evan Brady  |  1988-12-18  |  Cloud Cost Optimisation  |  Ruth Collins  |  2025-05-14  |  Cloud Suite
Liam Byrne  |  1990-2-24  |  Scaling Neo4j for Recommendations  |  Prof. Alan Shaw  |  2025-05-12  |  Graph Lab
Liam Byrne  |  1990-2-24  |  Customer 360 with SQL  |  Dr. Niamh Burke  |  2025-05-14  |  Main Hall
```

Figure 7 Details of attendees from company ID 2

A valid company ID is any number greater than 0.

The user is asked to enter a valid company ID until one has been entered

```
Enter Company ID : -1
Enter Company ID : asdf
Enter Company ID :
```

Figure 8 Invalid company IDs.

3.1.4.1 Error Conditions

The following error conditions must be handled as described, and allow the user another opportunity to enter a correct company ID.

- A valid company ID is entered, but that company doesn't exist:

```
Choice: 2

Enter Company ID : 99
Company with ID  99  doesn't exist
```

Figure 9 Non-existent company

- A valid company ID is entered and the company exists, but there are no attendees from this company for any sessions:

```
Choice: 2

Enter Company ID : 9
WindyDays  Attendees
No attendees found for  WindyDays
```

Figure 10 Company with no attendees

3.1.5 3 (Add New Attendee)

The user is asked to enter the following details for a new attendee:

- ID
- name
- DOB
- Gender
- ID attendee's company

```
Choice: 3

Add New Attendee
----------------
Attendee ID : 121
Name : Joe Kelly
DOB : 1970-02-18
Gender : Male
Company ID : 2

Attendee successfully added
```

Figure 11 New Attendee Details

When the attendee has been successfully added to the database the message "Attendee successfully added" should be shown, and the user returned to the main menu.

```
mysql> SELECT * FROM attendee WHERE attendeeID = 121;
+------------+--------------+-------------+----------------+-------------------+
| attendeeID | attendeeName | attendeeDOB | attendeeGender | attendeeCompanyID |
+------------+--------------+-------------+----------------+-------------------+
|        121 | Joe Kelly    | 1970-02-18  | Male           |                 2 |
+------------+--------------+-------------+----------------+-------------------+
1 row in set (0.00 sec)
```

Figure 12 New attendee added to database

3.1.5.1 Error Conditions

- If the Attendee ID already exists, the user should be informed via the following error message:

```
Choice: 3

Add New Attendee
----------------
Attendee ID : 101
Name : Joe Kelly
DOB : 1970-02-18
Gender : Male
Company ID : 2
*** ERROR *** Attendee ID: 101 already exists
```

Figure 13 Attendee ID already exists

- If an invalid gender is entered, the user should be informed via the following error

```
Choice: 3

Add New Attendee
----------------
Attendee ID : 122
Name : Mary McGovern
DOB : 1987-08-23
Gender : asdf
Company ID : 2
*** ERROR *** Gender must be Male/Female
```

Figure 14 Invalid gender

- If a non-existent Company ID is entered, the user should be informed via the following error message:

```
Choice: 3

Add New Attendee
----------------
Attendee ID : 122
Name : Mary McGovern
DOB : 1987-08-23
Gender : Female
Company ID : 999
*** ERROR *** Company ID: 999 does not exist
```

Figure 15 Invalid Company ID

- For any other input error e.g. invalid Attendee ID or DOB, the error reported by the database can be shown:

```
Choice: 3

Add New Attendee
----------------
Attendee ID : asdf
Name : Mary McGovern
DOB : 1987-08-23
Gender : Female
Company ID : 2
*** ERROR *** (1366, "Incorrect integer value: 'asdf' for column 'attendeeID' at row 1")
```

Figure 16 Invalid Attendee ID

```
Choice: 3

Add New Attendee
----------------
Attendee ID : 123
Name : Mary McGovern
DOB : asdf
Gender : Female
Company ID : 2
*** ERROR *** (1292, "Incorrect date value: 'asdf' for column 'attendeeDOB' at row 1")
```

Figure 17 Invalid DOB

3.1.6 4 (View Connected Attendees)

The user is asked to enter an attendee ID. The name of the attendee as well as the ID and name of all other attendees that have a *CONNECTED_TO* relationship (in either direction) to this attendee are shown.

```
Choice: 4

Enter Attendee ID : 101
Attendee Name:  Ava Murphy
--------------------
These attendees are connected:
107  |  Mia O'Brien
109  |  Ella Finn
111  |  Grace Power
```

Figure 18 Example of connected attendees

If an attendee does not exist in the Neo4j database, but does exist in the MySQL database, his/her name should be shown along with "No connections"

```
Choice: 4

Enter Attendee ID : 112
Attendee Name:  Daniel Quinn
--------------------
No connections
```

Figure 19 Attendee in MySQL but not Neo4j

If an attendee does not exist neither the Neo4j database, nor the MySQL database. The following message should be shown:

```
Enter Attendee ID : 999
*** ERROR *** Attendee does not exist
```

Figure 20 Attendee not in MySQL or Neo4j DBs

3.1.6.1 Error Conditions

The following error conditions must be handled as described, and allow the user another opportunity to enter a correct attendee ID.

- Non-numeric attendee ID is entered:

```
Choice: 4

Enter Attendee ID : asdf
*** ERROR *** Invalid attendee ID
```

Figure 21 Invalid attendee

3.1.7 5 (Add Attendee Connection)

The user is asked to enter 2 Attendee IDs.

```
Choice: 5

Enter Attendee 1 ID : 117
Enter Attendee 2 ID : 118
Attendee 117 is now connected to Attendee 118
```

Figure 22 Enter two attendee IDs

If these attendee IDs exist in the MySQL database, and neither actor has a *CONNECTED_TO* relationship in the Neo4j database, then a *CONNECTED_TO* relationship should be created between both nodes.

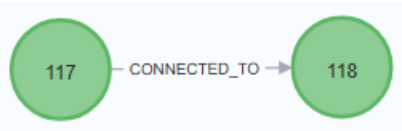

Figure 23 Attendee 117 is now CONNECTED_TO attendee 118 (and vice versa)

In this example, one of the attendees (112) exists in the MySQL database, but not in the Neo4j database:

```
Choice: 5

Enter Attendee 1 ID : 112
Enter Attendee 2 ID : 115
Attendee 112 is now connected to Attendee 115
```

Figure 24 One attendee not already in Neo4j DB

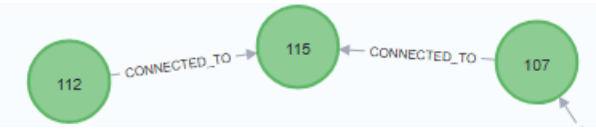

Figure 25 Attendee 112 is now CONNECTED_TO attendee 115

3.1.7.1 Error Conditions

The following error conditions must be handled as described, and allow the user another opportunity to enter correct attendee IDs.

- An attendee cannot be *CONNECTED_TO* him/herself:

```
Choice: 5

Enter Attendee 1 ID : 113
Enter Attendee 2 ID : 113
*** ERROR *** An attendee cannot connect to him/herself

Enter Attendee 1 ID :
```

Figure 26 Error Condition

- If an attendee is already *CONNECTED_TO* another attendee (regardless of direction of relationship) they cannot be connect again:

```
Choice: 5

Enter Attendee 1 ID : 113
Enter Attendee 2 ID : 105
*** ERROR *** These attendees are already connected

Enter Attendee 1 ID :
```

Figure 27 Error condition

- If an attendee doesn't exist (in the MySQL database) a node representing him/her should not be created in the Neo4j database.

```
Choice: 5

Enter Attendee 1 ID : 999
Enter Attendee 2 ID : 113
*** ERROR *** One or both attendee IDs do not exist

Enter Attendee 1 ID :
```

Figure 28 Error condition

- Non-numeric attendee IDs are entered:

```
Choice: 5

Enter Attendee 1 ID : eee
Enter Attendee 2 ID : 3k3
*** ERROR *** Attendee IDs must be numbers

Enter Attendee 1 ID :
```

Figure 29 Both actors already married

3.1.8 6 (View Rooms)

When this option is chosen the:

- Room ID
- Room name
- Capacity

of all rooms is shown:

```
Choice: 6
RoomID   | RoomName | Capacity
1  |  Main Hall  |  300
2  |  Graph Lab  |  120
3  |  Cloud Suite  |  180
4  |  Innovation Room  |  90
5  |  Workshop Studio  |  60
6  |  Executive Lounge  |  40
```

Figure 30 Rooms

Any new rooms manually added to the MySQL database, after this option has been chosen for the first time, should not be displayed until the user exits and restarts the application.

3.1.9 x (Exit Application)

The program terminates.

3.1.10 Anything Else

The menu is shown again.